EXP 4 Design of Low Pass Filter using Windowing Techniques

4..1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in Objective: MATLAB

clc; clear; close all;

%% --- Input Parameters ---fp = 1000;

% Passband cutoff (Hz)

fs = 4000;   % Sampling frequency (Hz)

N = 50;   % Filter order (even number preferred) wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---b_rect = fir1(N, wc, 'low', rectwin(N+1));

disp('--- Rectangular Window Coefficients ---'); disp(b_rect');

%% --- 2. FIR Low Pass using Hamming Window ---b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---'); disp(b_hamm');

%% --- 3. FIR Low Pass using Hanning Window ---b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---'); disp(b_hann');

%% --- Plot Frequency Responses ---figure;

freqz(b_rect,1);

title('Low-pass FIR using Rectangular Window'); figure;

freqz(b_hamm,1);

title('Low-pass FIR using Hamming Window'); figure;

freqz(b_hann,1);

title('Low-pass FIR using Hanning Window');

OUTPUT

--- Rectangular Window Coefficients ---0.0126

0

-0.0137

0

0.0150

0

-0.0166

0

0.0185

0

-0.0210

0

0.0242

0

-0.0286

0

0.0349

0

-0.0449

0

0.0629

0

-0.1048

0

0.3145

0.4940

0.3145

0

-0.1048

0

0.0629

0

-0.0449

0

0.0349

0

-0.0286

0

0.0242

0

-0.0210

0

0.0185

0

-0.0166

0

0.0150

0

-0.0137

0

0.0126

--- Hamming Window Coefficients ---0.0010

0

-0.0013

0

0.0021

0

-0.0034

0

0.0055

0

-0.0084

0

0.0125

0

-0.0181

0

0.0260

0

-0.0379

0

0.0580

0

-0.1026

0

0.3168

0.4995

0.3168

0

-0.1026

0

0.0580

0

-0.0379

0

0.0260

0

-0.0181

0

0.0125

0

-0.0084

0

0.0055

0

-0.0034

0

0.0021

0

-0.0013

0

0.0010

--- Hanning Window Coefficients ---0.0000

0

-0.0004

0

0.0013

0

-0.0028

0

0.0050

0

-0.0081

0

0.0122

0

-0.0179

0

0.0259

0

-0.0378

0

0.0580

0

-0.1027

0

0.3172

0.5000

0.3172

0

-0.1027

0

0.0580

0

-0.0378

0

0.0259

0

-0.0179

0

0.0122

0

-0.0081

0

0.0050

0

-0.0028

0

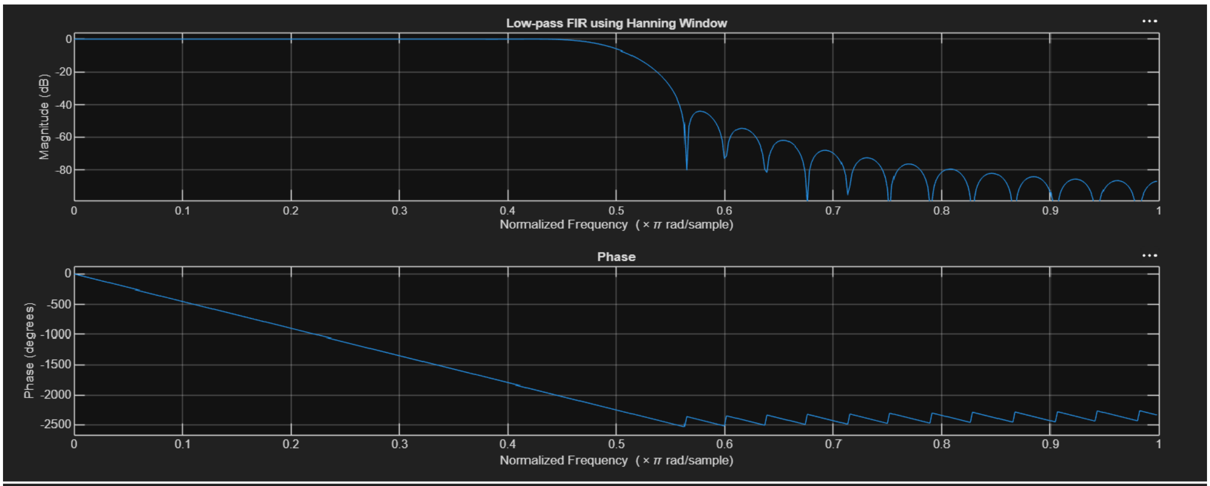
0.0013

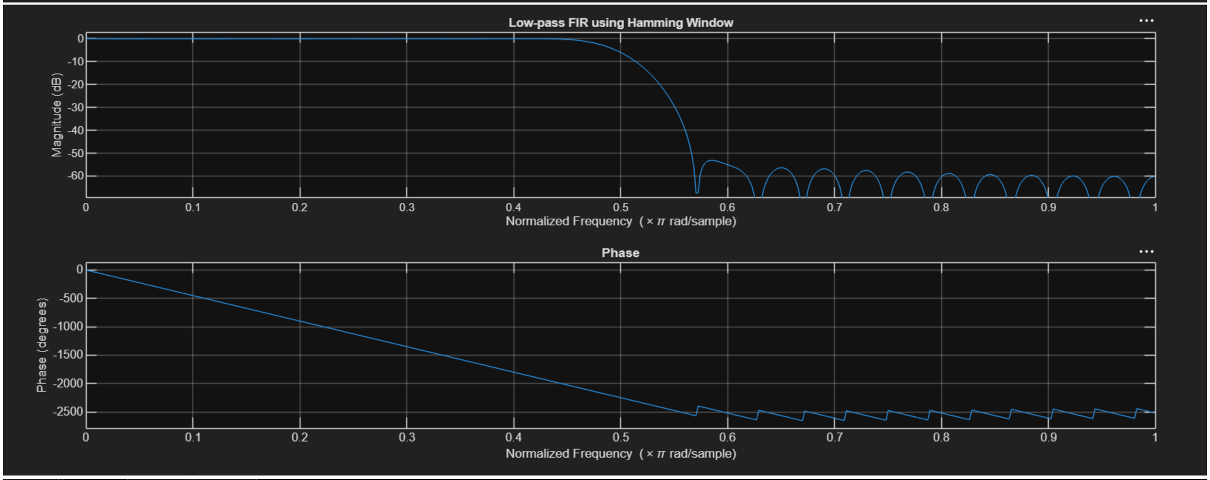
0

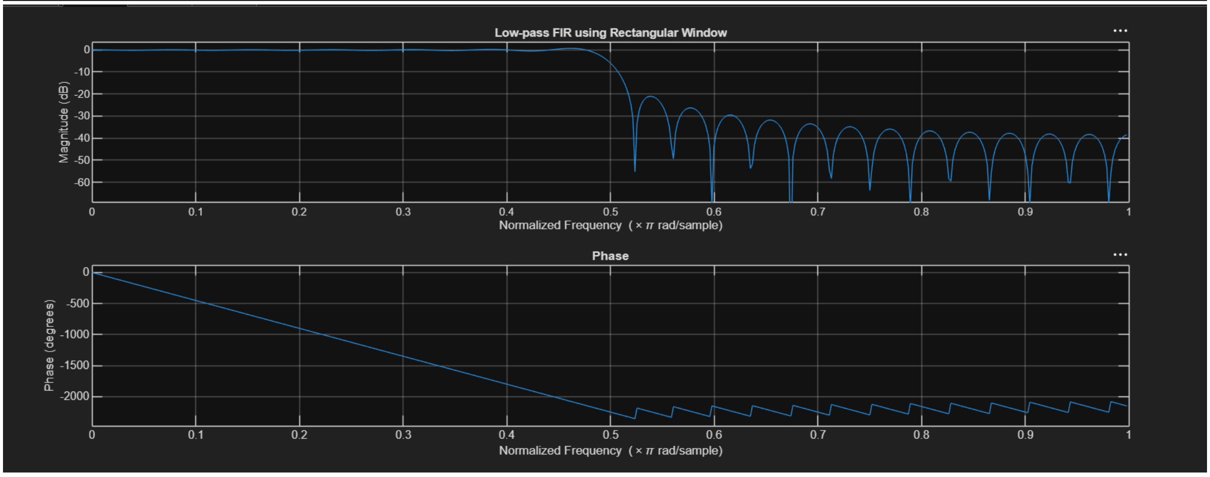
-0.0004

0

0.0000


Low-pass FIR using Hanning Window
Magnitude (dB)
0
-20
-40
-60
-80
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-500
-1000
-1500
-2000
-2500
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)


Low-pass FIR using Hamming Window
Magnitude (dB)
0
-10
-20
-30
-40
-50
-60
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-500
-1000
-1500
-2000
-2500
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)


Low-pass FIR using Rectangular Window
Magnitude (dB)
0
-10
-20
-30
-40
-50
-60
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-500
-1000
-1500
-2000
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)